The Islamia University of Bahawalpur
Department of Information Technology

INFT-4041 · BSIT · 5th Semester · Fall 2026

E-Commerce Application Development

Chapter 4

Planning an E-Commerce Framework

Part I — E-Commerce and Its Models

Taught in	Week 2
Laboratory	Three shapes compared; your requirements and your decision
Assessment	—

This is a class extract. It is *Chapter 4* of the unified handout, printed on its own so that a week's reading is a week's reading. Page numbers restart at 1 and are local to this extract; chapter, figure and table numbers are the ones the full volume gives them, so a reference you follow here lands in the same place there.

Cross-references to other chapters print as their numbers in the full volume. For the whole course in one file — front matter, the HEC tracing table, all twenty-three chapters and all seven appendices — see `ECAD_Fall2026_Handout.pdf`.

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Chapter 4

Planning an E-Commerce Framework

Learning Outcomes

By the end of this chapter you will be able to:

- **Distinguish** the four ways a shop can be obtained, and **state** what each costs in money, time and control.
- **Write** requirements for a shop in a form that discriminates between those four rather than describing all of them equally.
- **Identify** the small number of requirements that actually decide the choice, and **explain** why most requirements do not.
- **Compute** a total cost of ownership over three years, including the costs that do not appear on a price list.
- **Defend** a build-buy-or-extend decision against the obvious objection to it.

Before You Start

Chapter 2 for the business and revenue model, and its laboratory — this chapter turns that one-page brief into a decision. Chapter 3 if your venture is a marketplace rather than a shop.

Key Terms in This Chapter

requirement • functional requirement • non-functional requirement • SaaS • hosted platform • self-hosted • open source • bespoke development • total cost of ownership • vendor lock-in • extensibility • data portability • decision matrix • minimum viable product

4.1 The Question This Chapter Answers

You have a business model, a revenue model and some idea of what you are selling. There is now exactly one decision to make before any of the rest of this book applies, and it is this: *where does the software come from?*

There are four answers, and the whole of Parts II to V is a consequence of which one you pick.

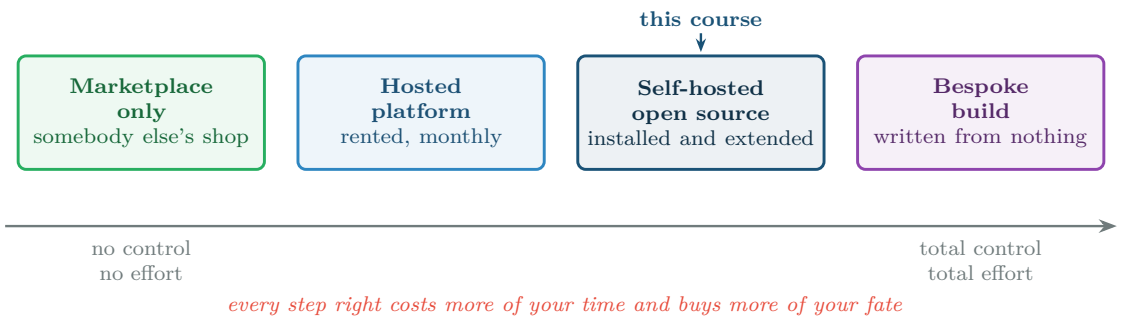


Figure 4.1: The four ways to obtain a shop. They are a spectrum, not a menu: each step to the right buys control and costs effort, and the right answer depends on which of the two you are shorter of.

4.2 The Four Options, Honestly

Option	What it is good at	What it costs you
Marketplace only	Customers on day one. No software, no hosting, no payment integration. Genuinely the right answer for many small sellers	You perform none of the four functions (Chapter 1). You do not own the customer relationship, cannot email them, and can be delisted without appeal. The commission is the visible cost and the smallest one
Hosted platform	Running in an afternoon, someone else patches it, payment and shipping already integrated	A monthly fee that scales with your success, a transaction cut on top, and a hard ceiling: when the platform cannot do something, it cannot do it. Getting your data out ranges from awkward to impossible
Self-hosted open source	You can change anything, including the things the vendor never anticipated. No per-sale cut. The data is in your database	You are now responsible for hosting, backups, updates and *security* — an unpatched shop is a liability, not an inconvenience. Needs someone who can do Chapter 20
Bespoke build	Fits the business exactly, because it was built for it. The only option when the business is genuinely unusual	Twelve to eighteen months before you sell anything, and you will reimplement baskets, tax tables and payment retries that have been solved for twenty years — badly, because you will meet each edge case for the first time in production

⚠ The outline this course follows assumes the fourth option

The HEC outline for this course is, clause for clause, the contents page of a 2010 book about writing a shopping cart from scratch in PHP 5 — see the Course Specification. Taken literally, it asks a 5th-semester class to choose *bespoke build*.

That was defensible in 2010 and is not now, for a reason this chapter can state precisely: the work has moved. In 2010, writing a basket and a checkout was most of the job. Today a mature, free, auditable implementation of both is one command away, and the job that remains — modelling the catalogue, configuring tax and shipping, integrating payment, securing the result, extending what the platform cannot do — is both larger and more representative of what an IT graduate is actually hired to do.

So this course teaches every *topic* the outline lists, and reaches them by the third route rather than the fourth. You should be able to defend that choice, not merely inherit it, which is what this chapter is for — and Chapter 19 is where the "extend" in "self-hosted and extend" stops being a word and becomes code you write.

4.3 Requirements That Actually Discriminate

Students write requirements like "the system shall allow customers to add products to a basket". Every one of the four options does that. A requirement that all four satisfy carries no information and cannot help you choose.

Functional and Non-Functional Requirements

A **functional requirement** says what the system must *do*: accept a voucher code, calculate tax by province, email an invoice.

A **non-functional requirement** says how well, under what constraints: how fast, how many at once, how available, how secure, how maintainable, by whom.

For the build-buy-extend decision the useful requirements are nearly all non-functional or unusual-functional. The ordinary functional ones are table stakes — everything does them — and listing them at length is the most common way a requirements document ends up being thirty pages that decide nothing.

The five questions that decide it

In practice the choice turns on a handful of questions. Ask these first, and most of the rest is detail.

1. **Is there anything genuinely unusual about how you price, sell or deliver?** Per-customer negotiated pricing, made-to-measure items, rental with a return date, services booked by time slot. If yes, every option except the last two gets harder fast.
2. **Who will maintain it in two years?** Not who will build it. A self-hosted shop with nobody to patch it becomes a breach (Chapter 16). If the honest answer is nobody, choose a hosted platform and accept the ceiling.
3. **What volume, and how uneven?** Thirty orders a day and three thousand are different systems; so are steady traffic and traffic that multiplies tenfold for one week a year.
4. **How much does leaving cost?** Can you export your products, customers and order history in a usable form? Ask it on day one, because the answer never improves.
5. **What is the budget, over three years, including somebody's time?** Not the setup cost. See below.

4.4 The Cost That Is Not on the Price List

The commonest error here is comparing a monthly fee against "free" and concluding that open source is cheaper. Both numbers are wrong.

Worked Example: Three years of a small shop, two ways

A shop expecting **Rs. 800,000 of sales a month**, growing slowly. Compare a hosted platform against self-hosted open source over three years.

Hosted platform.

- Subscription: \$79/month ≈ Rs. 22,000. Over 36 months: **Rs. 792,000**.
- Platform transaction fee, 0.5% of sales — charged on top of the gateway's own fee: $0.005 \times 800{,}000 \times 36 =$ **Rs. 144,000**.
- Two paid add-ons at \$15/month each ≈ Rs. 8,400/month: **Rs. 302,400**.
- Theme and setup, one-off: **Rs. 60,000**.
- Maintenance labour: near zero; the vendor patches.
- **Total: Rs. 1,298,400.**

Self-hosted open source.

- Licence: **Rs. 0**. This is the number people stop at.
- Hosting, a modest virtual server plus backups, Rs. 9,000/month: **Rs. 324,000**.
- Setup and theming, 70 hours of somebody at Rs. 2,500/hour: **Rs. 175,000**.
- Two custom modules the platform does not ship, 40 hours: **Rs. 100,000**.
- **Maintenance: 4 hours a month** — updates, patches, backups checked, the occasional breakage. $4 \times 36 \times 2{,}500 =$ **Rs. 360,000**.
- **Total: Rs. 959,000.**

The result. Self-hosting wins by Rs. 339,400 over three years — about 26%. But look at what decided it: **maintenance labour is Rs. 360,000**, the single largest line in the self-hosted column and the one most often entered as zero. Set it to zero and self-hosting appears to win by 87%. Double it, to eight hours a month because nobody on staff really knows the platform, and the two options are level.

The lesson. This calculation is not really about software prices. It is a question about whether you have, and will keep, somebody who can maintain a server — and if the answer is no, the hosted option is cheaper at any subscription price, because the alternative is not a cheaper shop but an unpatched one.

⚠ Two costs that appear in neither column

Vendor lock-in is not a line item until the day you want to leave, at which point it is the whole bill. Before committing, actually export your catalogue and customers from a trial account and look at what comes out. "We support CSV export" and "you can reconstruct your shop from this" are different claims.

The ceiling has no price at all. It is the feature you cannot have, discovered in month fourteen, when the business depends on it. The hosted option's real cost is the probability that this happens multiplied by what it does to you — which is why question 1 above is first.

4.5 Making the Decision Defensible

Write it down as a matrix: criteria in rows, weights, options scored. Not because the arithmetic is wise — it is easy to fit the weights to the answer you wanted — but because it forces the weights into the open where somebody can disagree with them.

</> A decision matrix for the shop in this course

Criteria weighted out of 100, each option scored 1–5.

Criterion	Weight	Hosted	Self-host	Bespoke
Students must see and change real code	30	1	5	5
Working shop within three weeks	25	5	4	1
Zero licence or subscription cost	15	2	5	5
Runs identically on every laptop	15	3	5	4
Covers the HEC clause list directly	15	3	5	4

Weighted total	**100**	**2.75**	**4.75**	**3.70**

Read the weights, not the totals. The first row is worth 30 and it is where hosted loses the contest — a course whose students cannot read the code is not this course. Change that weight to 5, as a small shopkeeper would, and hosted wins outright. The matrix did not make the decision; it made the decision *visible*, which is all a matrix is for.

Common Pitfalls

- **Requirements that every option satisfies.** "Customers can add to a basket" discriminates between nothing.
- **Entering maintenance as zero.** It was the largest single line in the worked example, and leaving it out reverses the answer.
- **Comparing setup cost instead of three-year cost.** Subscriptions and labour both accumulate; neither appears in a setup quote.
- **Choosing bespoke because the platform is "bloated".** You will reimplement the parts you called bloat, in production, one incident at a time.
- **Choosing self-hosted with nobody to maintain it.** This is not a cheaper shop. It is a shop that will be compromised, and Chapter 16 shows how quickly.
- **Not testing the export before committing.** Lock-in costs nothing until it costs everything.
- **Weighting the matrix after seeing the totals.** Everyone is tempted; write the weights down first and date them.

Try It Yourself: Your requirements, and your decision

The last analysis laboratory. Chapter 5 installs software.

Using the venture from Chapter 2's laboratory, produce a two-page document:

1. **Twelve requirements**, each labelled functional or non-functional, each numbered so you can refer to it later. At least four must be non-functional, with a number attached: not "must be fast" but "a category page must render in under one second with 5,000 products". You will test some of these in Chapter 6 and Chapter 20, so make them checkable.
2. **Mark the discriminating ones.** Go through your twelve and strike out every requirement that all four options satisfy. Most students strike out eight or nine. What is left is your real requirements document.
3. **Answer the five questions** from this chapter, in writing, including the honest answer to "who maintains this in two years?"
4. **A three-year total cost of ownership**, two columns, following the worked example. Mark every figure you guessed, and state the one number the answer is most sensitive to.
5. **A decision matrix**, weights written before scores, and a two-sentence conclusion.
6. **The strongest objection to your decision**, and your answer to it. If you cannot state a real objection, you have not understood the option you rejected.

What to notice. Your decision does not have to be the one this course makes. A student whose venture is three products sold through Instagram should conclude "marketplace only" and say so — and will still build the shop in Parts II–V, because the course has to teach the general case. Being right about your own venture and building something else deliberately is a perfectly coherent position, and a more useful one than pretending.

Checkpoint — Test Yourself

1. Classify each as functional or non-functional, and say which of them discriminates between the four options: (a) customers can reset a forgotten password; (b) prices are agreed per customer and may differ for the same product; (c) the shop stays up during a sale that brings twenty times normal traffic.
2. A shop compares Rs. 22,000/month hosted against "free" open source and concludes open source saves Rs. 792,000 over three years. Name the three largest costs they have omitted.
3. Give a specific business for which this course's choice — self-hosted open source — would be the wrong answer, and say which option you would choose instead and why.

Chapter Summary

- A shop can be obtained four ways: marketplace only, hosted platform, self-hosted open source, or bespoke build. They form a spectrum trading control against effort.
- Requirements that every option satisfies cannot help you choose. Strike them out; what remains is the document that matters.
- Five questions decide most cases: anything unusual about pricing or delivery, who maintains it in two years, volume and its unevenness, the cost of leaving, and three-year budget including labour.
- Compare total cost of ownership over three years, not setup cost. In the worked example maintenance labour was the largest self-hosted line, and entering it as zero reverses the conclusion.
- Vendor lock-in and the feature ceiling appear in no price list and are often the real cost of the hosted option.
- A decision matrix does not make the decision; it makes the weights visible so somebody can argue with them. Write the weights first.
- This course chooses self-hosted open source and teaches every clause of the outline through it, rather than writing a cart from scratch as the 2010 book behind the outline assumed. You should be able to defend that, including knowing when it is the wrong answer.

Review Questions

1. (MCQ) Which is a non-functional requirement? (a) The shop emails an invoice after each order; (b) the checkout completes in under two seconds at 200 concurrent users; (c) customers can apply a voucher code; (d) administrators can issue refunds.

2. (MCQ) In the worked example, the largest single cost in the self-hosted column was: (a) hosting; (b) setup and theming; (c) custom modules; (d) maintenance labour.

3. (Short) Explain why a requirement satisfied by all four options is useless for this decision, and give an example of one.

4. (Short) What is vendor lock-in, and what concrete test would you run during a free trial to measure it?

5. (Short) Give the strongest argument *against* the choice this course makes, and then answer it.

6. (Applied) A tailoring business takes measurements in the shop, makes garments to order over two weeks, and prices each job individually after seeing the cloth. Work through the five questions and recommend one of the four options, justifying it. Identify the one requirement that makes this case different from a normal shop.

7. **(Applied)** Build a three-year total cost of ownership for a shop doing Rs. 3,000,000 a month — nearly four times the worked example — on both options, keeping all other assumptions. State which option wins, and identify the single assumption that, if changed, would reverse it.